Chapter 3

Bipolar junction transistors (BJTs)

Transistor construction

Transistor is a three terminal semiconductor device. These are:

- Emitter-heavily doped
- Base-lightly doped
- Collector-moderately doped

E	B	C	=	N	P	N	or	P	N	P

The outer layers have width much greater than the sandwiched p- or n-type material.

The arrow shows the direction of emitter current (conventional flow) through the device.

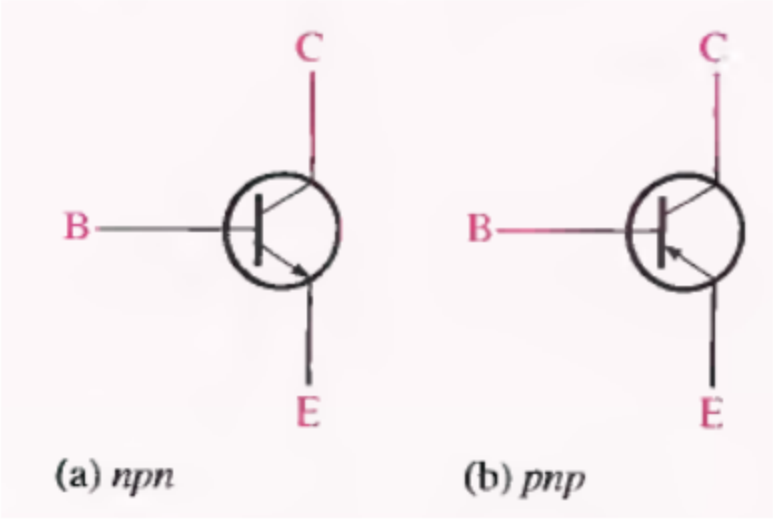


Fig. Standard BJT symbols

Types of configuration

	Common base	Common emitter	Common collector
Input	Emitter	Base	Base
Output	Collector	Collector	Emitter

- Transistor to operate properly as an amplifier, the two pn-junctions must be correctly biased with external DC voltages.
- Base-emitter junction forward biased and Base-collector junction reverse biased.

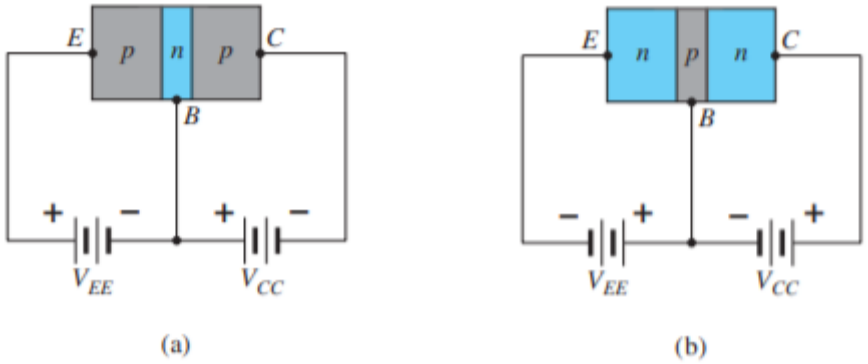
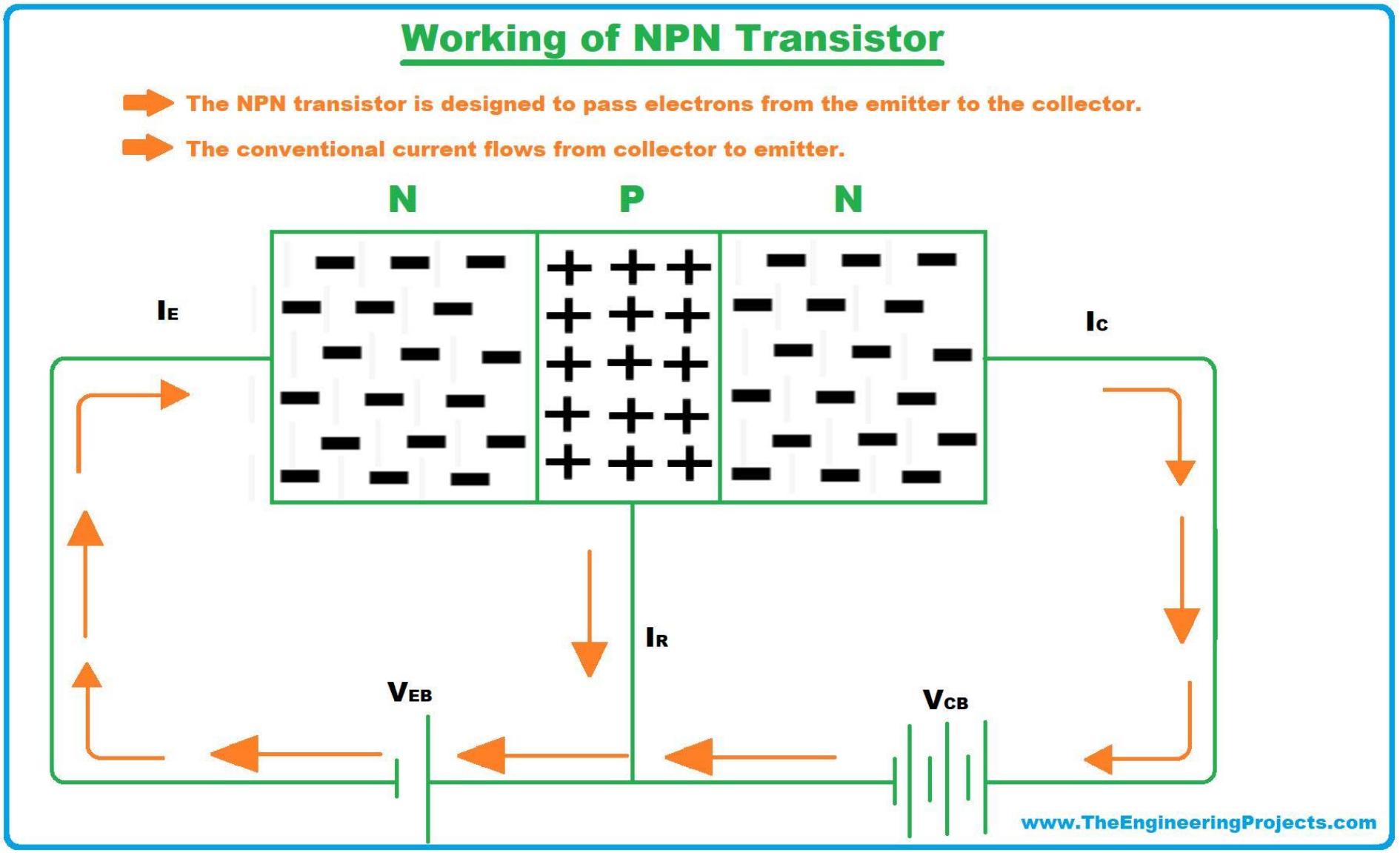


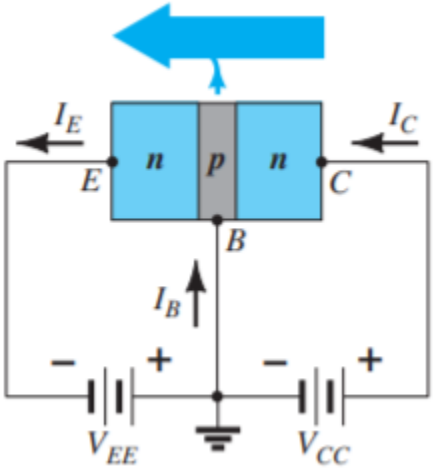
Fig. One p–n junction of a transistor is reverse-biased, whereas the other is forward-biased.

Transistor operation



Transistor operation

- The emitter base junction is forward biased by the dc source labeled Vee.
- Negative terminal is connected to n-side.



In N-type material

✓ electrons are the majority and

✓ holes are the minority

In p-type material

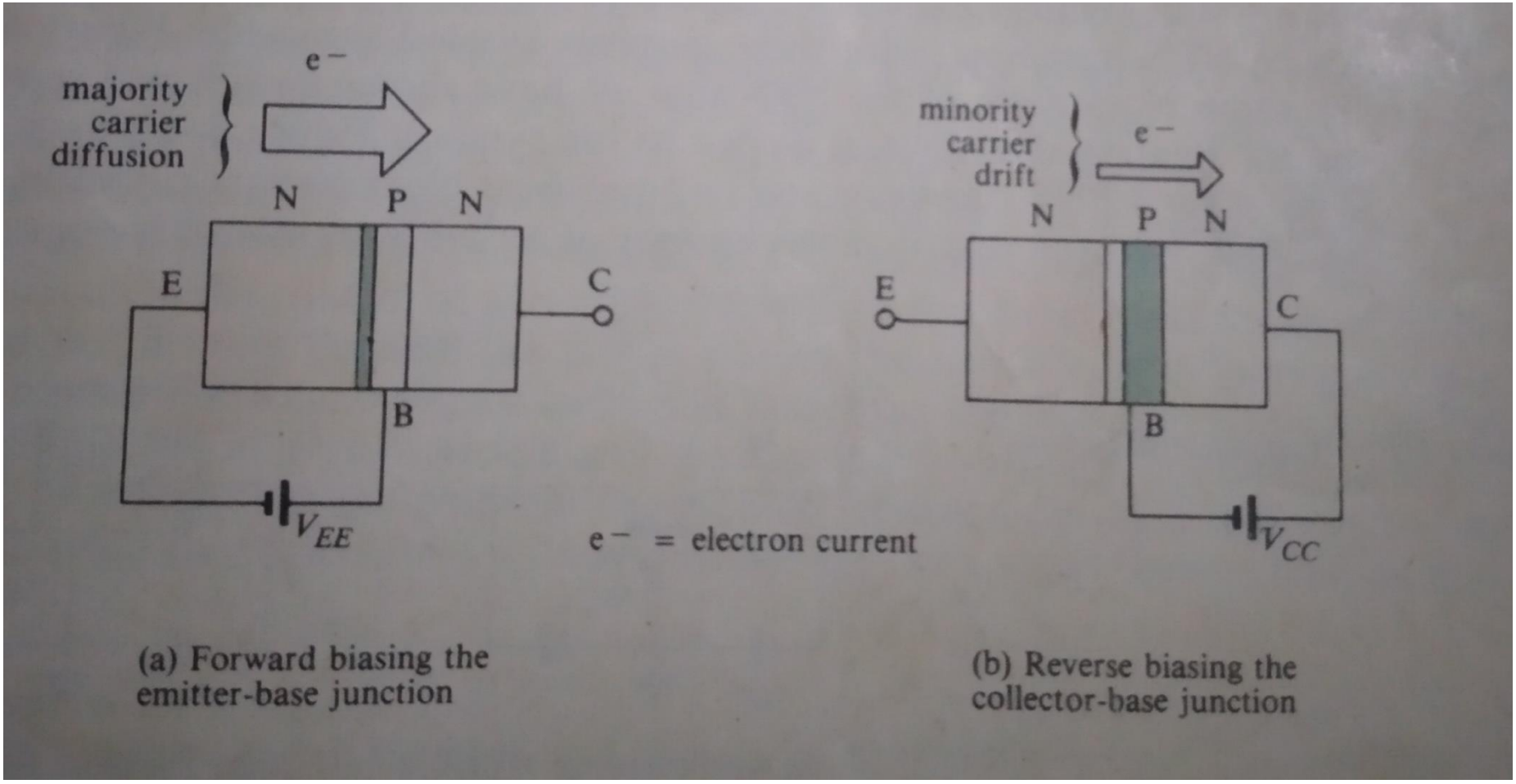
- ✓ electrons are the minority and
- ✓ holes are the majority

- A larger number of majority carriers (electrons) will diffuse across the forward biased np-junction in to the p-type material.

  -from emitter to base(n to p)

- The depletion region at this junction is made narrowed by a forward bias.

- When the majority electrons diffuse in to the base (p-type), they become minority carriers in that p-type region.
- It is said to be minority carriers have been injected into the base.



(a) Forward biasing the emitter-base junction

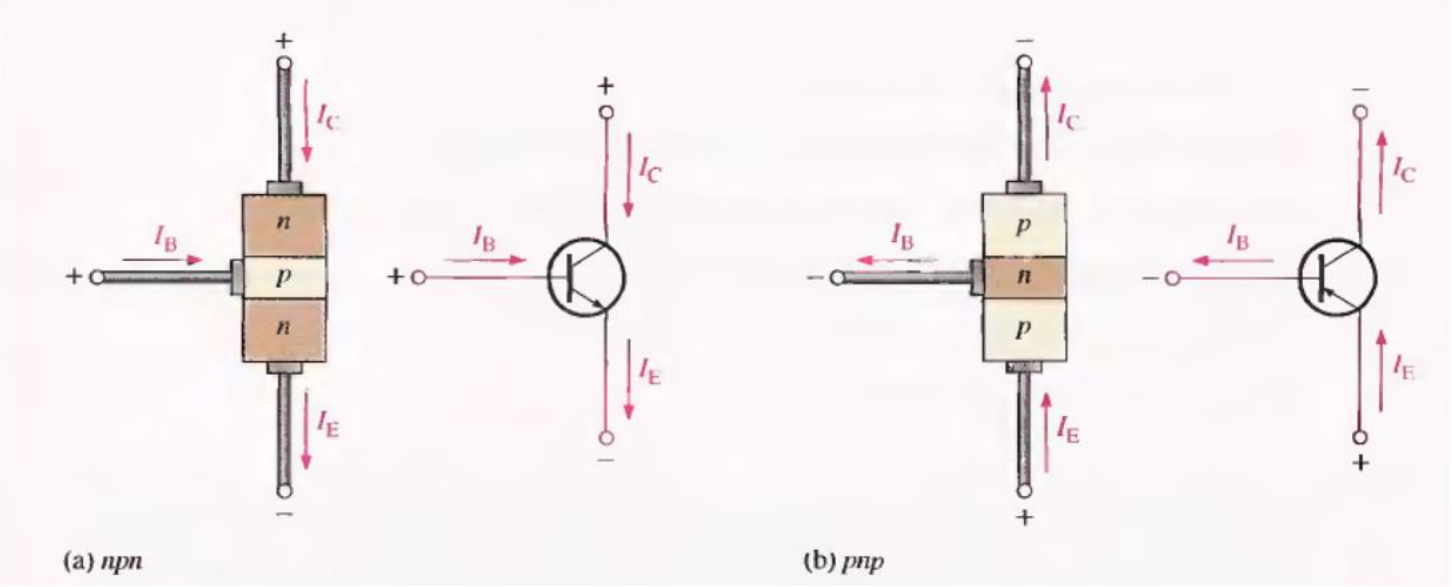
(b) Reverse biasing the collector-base junction

- **Collector base junction is reverse biased by the dc-source labeled Vcc.**
- **Positive terminal of Vcc is connected to the n-type collector.**
- **As a result, depletion region at this junction is widened and the only current that flows through base to collector is due to minority carriers (electrons) crossing the junction from the p-type base.**
- **Minority carriers crosses the reverse biased junction under the influence of electric field. (not by diffusion)**
- **Emitter is negative with respect to base and**
- **Collector is positive with respect to base.**
- **The base region is very thin and lightly doped.**

(there are few holes in it)

- Very few 2% of electrons injected in to the base from the emitter combine with holes and most of these electrons injected in to the base cross in to the collector.
- For each electron that combines with a hole, an electron leaves the base region via the base terminal.
- These action creates a very small base current I_B.

Transistor currents



$$I_E = I_B + I_C$$

TRANSISTOR CHARACTERISTICS AND PARAMETERS

DC Beta (β_{DC}) and DC Alpha (α_{DC})

The ratio of the dc collector current (I_C) to the dc base current (I_B) is the dc beta (β_{DC}), which is the DC current gain of a transistor.

$$\beta_{dc} = \frac{I_C}{I_B}$$

The ratio of the dc collector current (I_C) to the dc emitter current (I_E) is the dc alpha (α_{DC}).

$$\alpha_{DC} = \frac{I_C}{I_E}$$

EX-

Determine β_{DC} and I_E for a transistor where $I_B = 50\mu A$ and $I_C = 3.65$ mA.

Current and voltage analysis (CE)

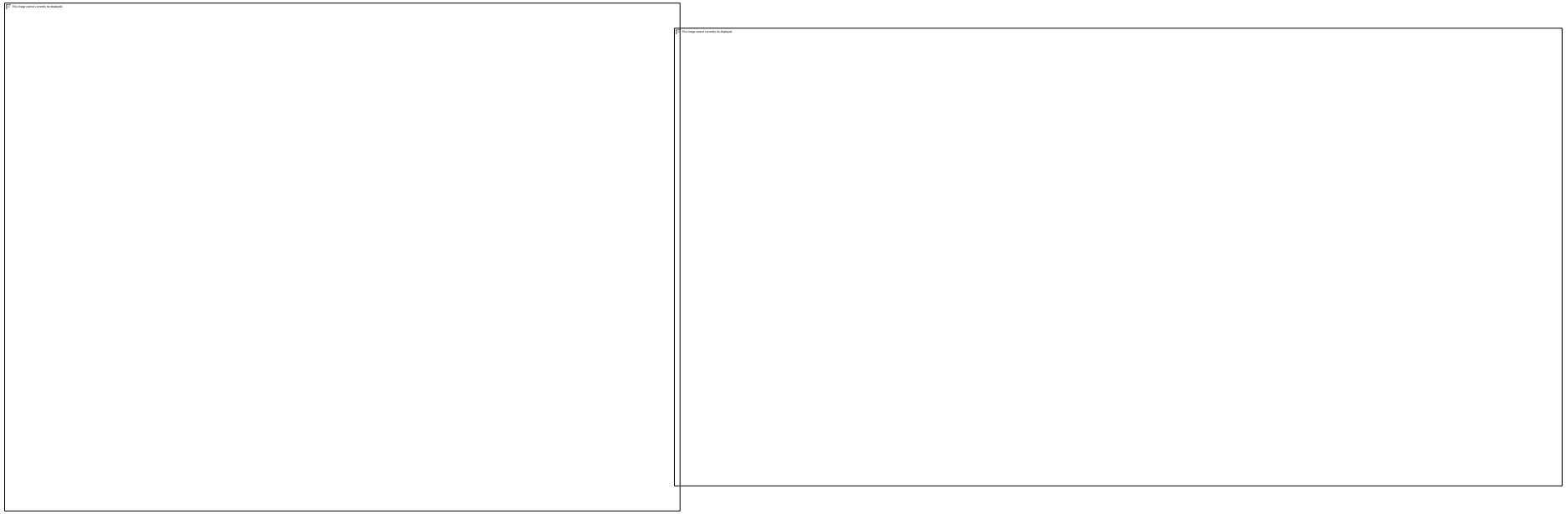


Fig. Transistor current and voltages

- V_{BB} forward-biases the base-emitter junction, and V_{cc} reverse-biases the base-collector junction.
- When the base-emitter junction is forward-biased, it is like a forward-biased diode and has a nominal forward voltage drop of $V_{BE} = 0.7$ V.

By Kirchhoff's voltage law, the voltage across R_B is

$$V_{RB} = V_{BB} - V_{BE}$$

Also, by Ohm's law,

$$V_{RB} = I_B R_B$$

Substituting for V_{RB} yields

$$I_B R_B = V_{BB} - V_{BE}$$

Solving for I_B ,

$$I_B = \frac{V_{BB} - V_{BE}}{R_B}$$

$$V_{RC} = I_C R_C$$

where $I_C = \beta_{DC} I_B$

The voltage at the collector with respect to the grounded emitter is

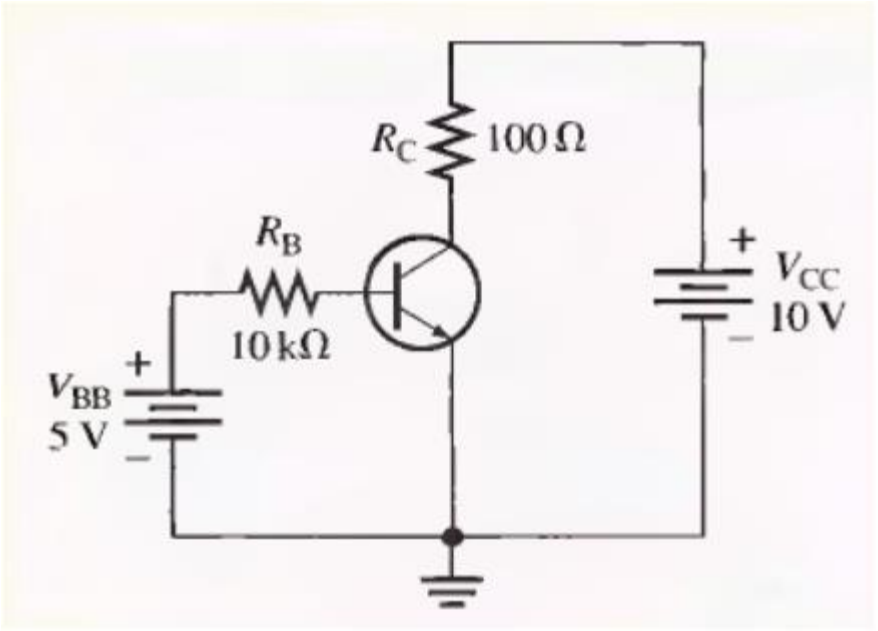
$$V_{CE} = V_{CC} - V_{R_C}$$

The voltage across the reverse-biased collector-base junction is

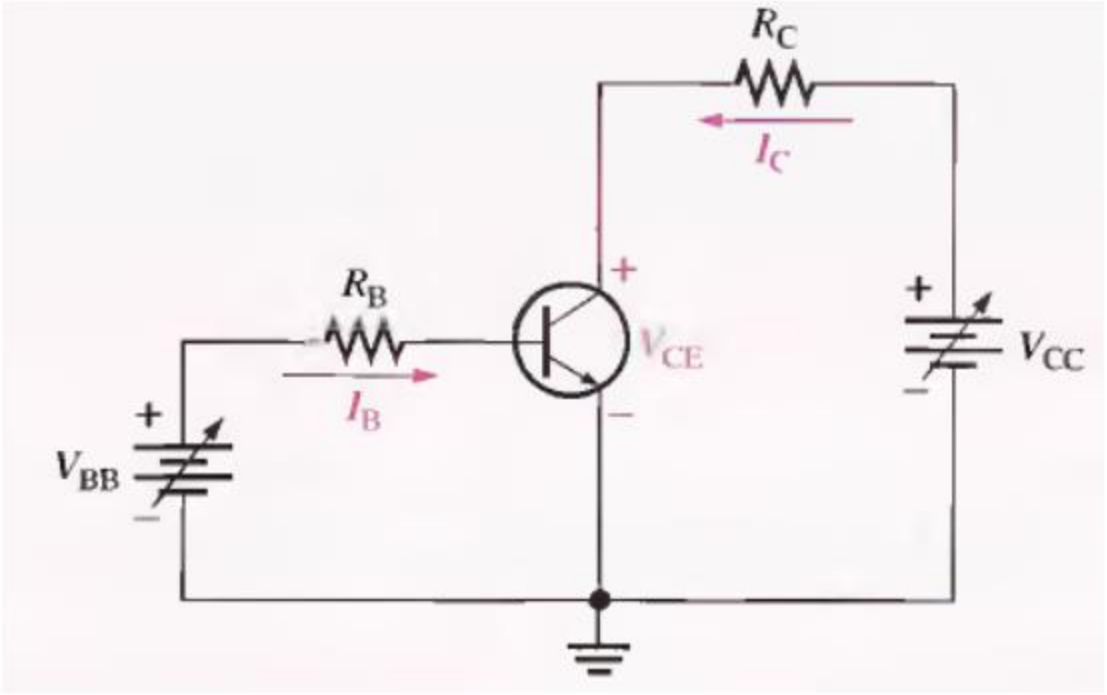
$$V_{CB} = V_{CE} - V_{BE}$$

Ex-

Determine $I_B, I_C, I_E, V_{BE}, V_{CE}$ *and* V_{CB} in the circuit of Fig. below. The transistor has a β_{DC} =150.



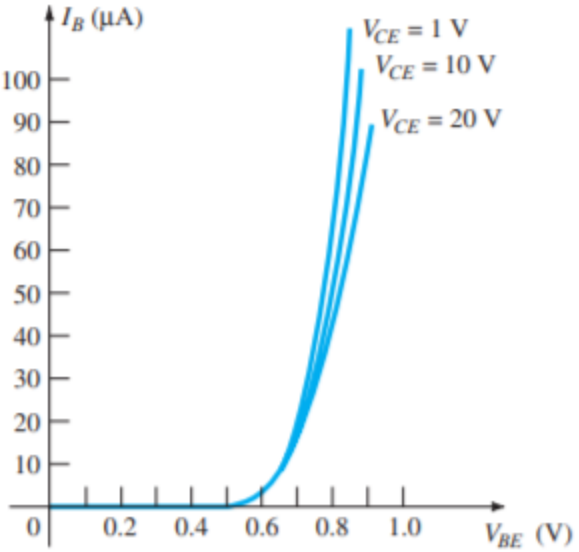
Common emitter input characteristics (base characteristics)



- The input to transistor in common emitter configuration is across the base to emitter junction.
- The CE input characteristics resemble a family of forward biased diode curves.

$\uparrow V_{CE} = \downarrow I_B$

$\uparrow V_{BE} = \uparrow I_B$



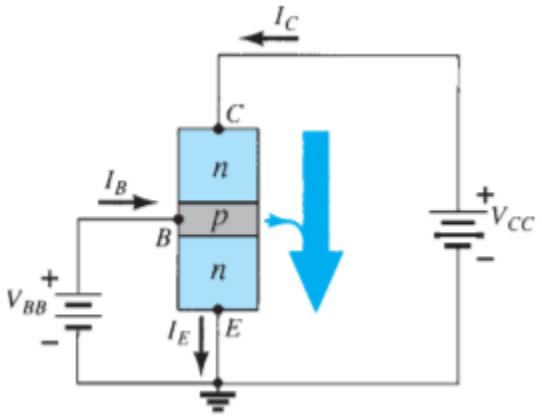


Fig. common emitter i/p characteristics

- I_B increases as V_{CE} decreases, for a fixed value of V_{BE}.
- A large value of V_{CE} results in large reverse bias of the collector base junction, which widens the depletion region and makes the base smaller and this results a reduction in base current.

Common emitter output characteristics(Collector Characteristic)

- Common emitter output characteristics show how the collector current, I_C varies with the collector-to-emitter voltage, V_{CE} for specified values of base current, I_B.

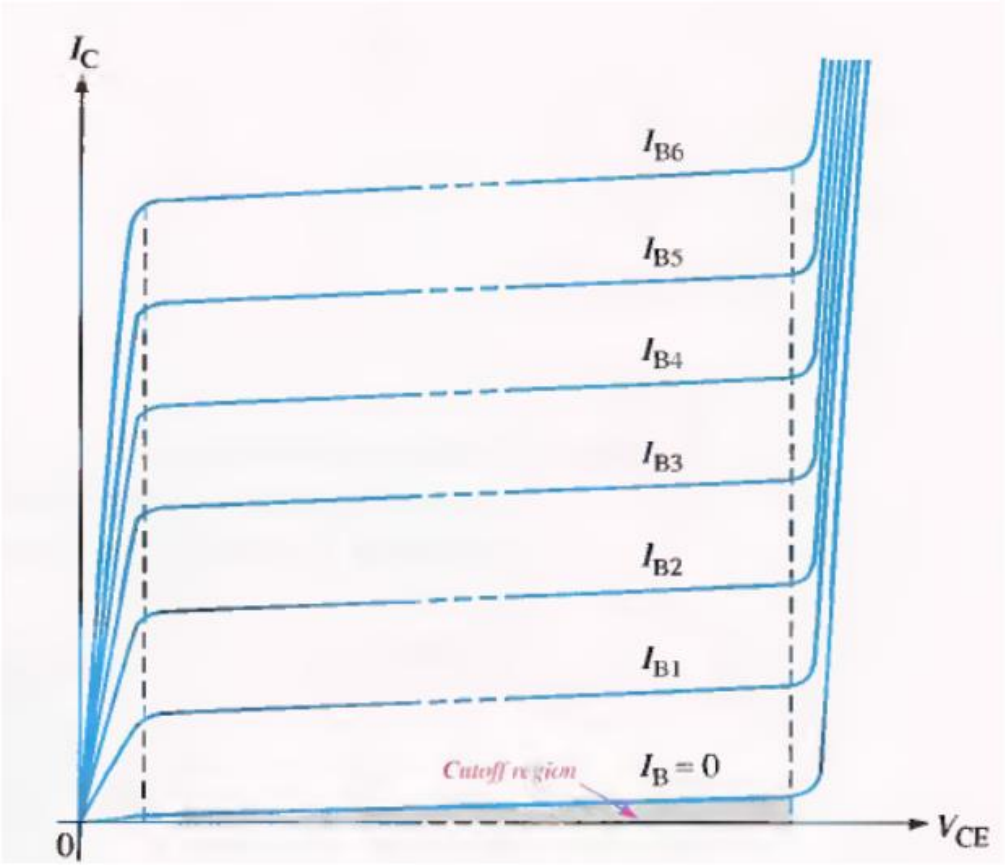
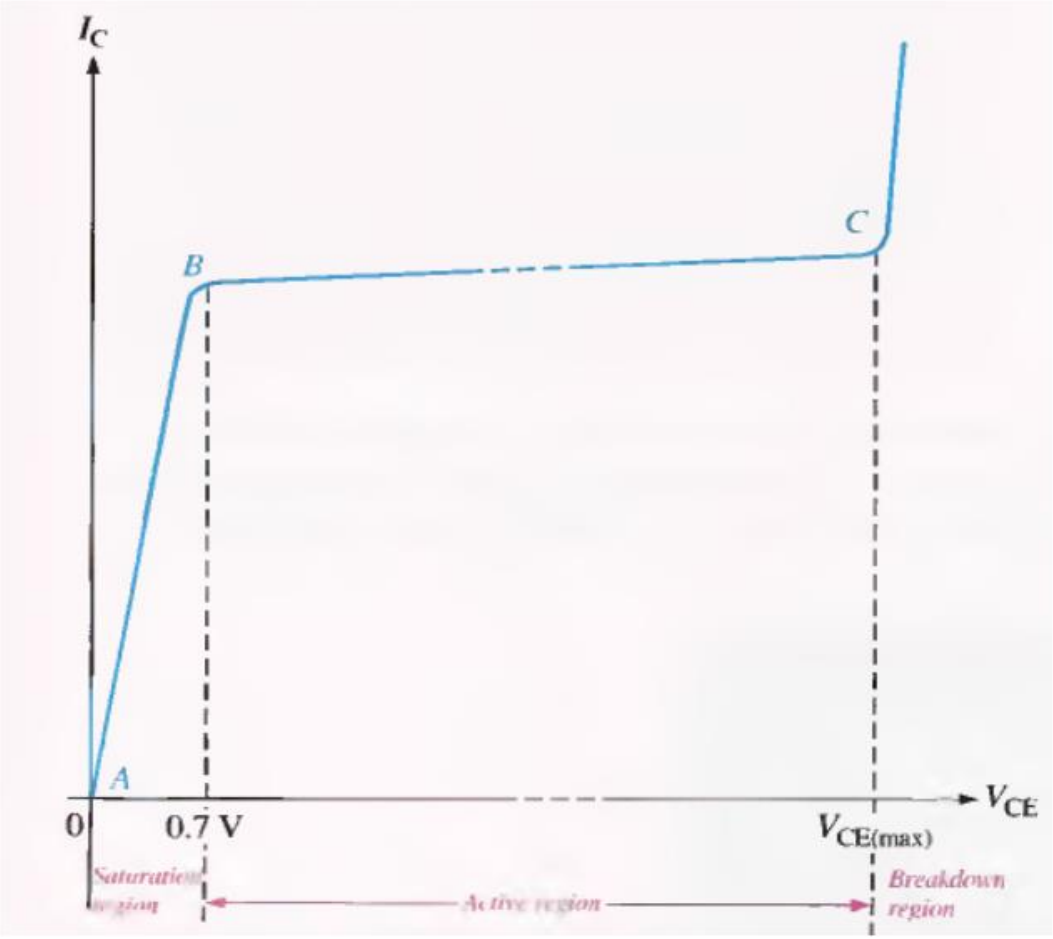


Fig. Family of I_C versus V_{CE} curves for several values of I_B ($I_{B1} < I_{B2} < I_{B3}$. etc.)

- **The region where the curves are approximately horizontal is the active region of the CE configuration.**
- **In this region β is essentially constant, but increase somewhat with V_{CE}.**
- **Collector current in CE configuration will increase rapidly if V_{CB} is permitted to become large.**
- **When both base-emitter junction and base-collector junctions are forward biased, the transistor is in the saturation region of its operation.**

Point A to B. ($V_{CE} < 0.7V$)

- **When base-emitter junction is forward biased and base-collector junction is reverse biased , the transistor is in the active or linear region.**



Point B to C. $(I_C = \beta I_B)$

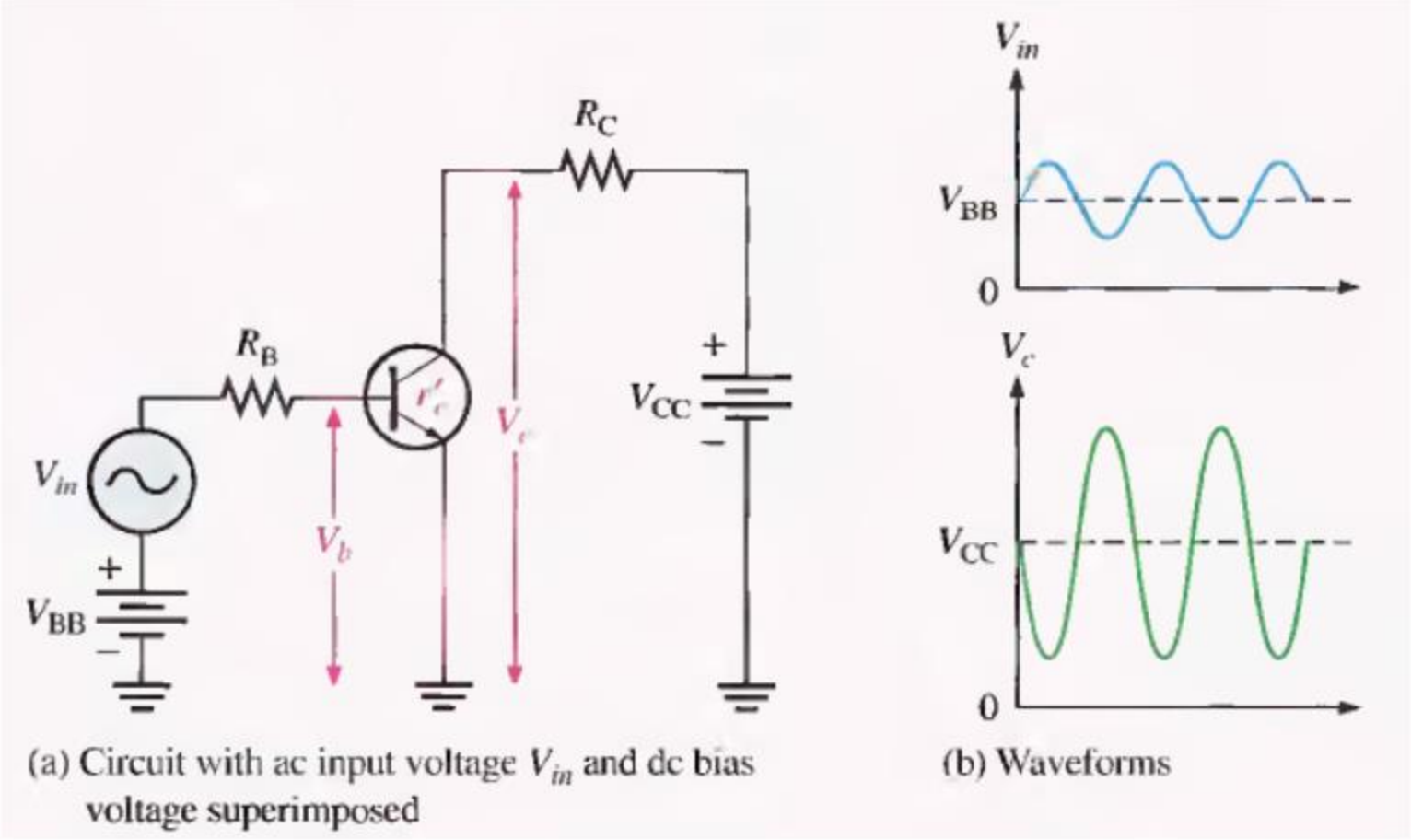
Fig. I_C versus V_{CE} curve for one value of I_B

- **When V_{CE} reaches a sufficiently high voltage, the reverse biased base-collector junction goes into break down and collector current increases rapidly and the transistor is said to be at its breakdown region.**

  At right of point C.

- **When $I_B = 0$, the transistor is in the cutoff region although there is a very small collector leakage current as indicated.**
- **In cut-off region, both BE-junction and BC-junctions are reversely biased.**

THE TRANSISTOR AS AN AMPLIFIER



(a) Circuit with ac input voltage V_{in} and dc bias voltage superimposed

(b) Waveforms

- **The ac input voltage produces an ac base current, which results in a much larger ac collector current.**
- **The ac collector current produces an ac voltage across R_C, thus producing an amplified but inverted wave form as shown in above figure.**
- **The forward biased, base-emitter junction presents a very low resistance (r_e) to the ac signal.**
- **Ac collector voltage, v_c, equals to the ac voltage drop across R_c.**

$$v_c = I_c R_c \approx I_e R_c$$

$$v_b = v_i - I_b R_b = I_e r_e$$

The ratio of v_c *to* v_b *is the ac voltage gain,* A_v.

$$A_v = \frac{v_c}{v_b} = \frac{I_e R_c}{I_e r_e} \approx \frac{R_c}{r_e}$$

R_c is always considerably larger than r_e → $v_o > v_i$

Transistor as switch

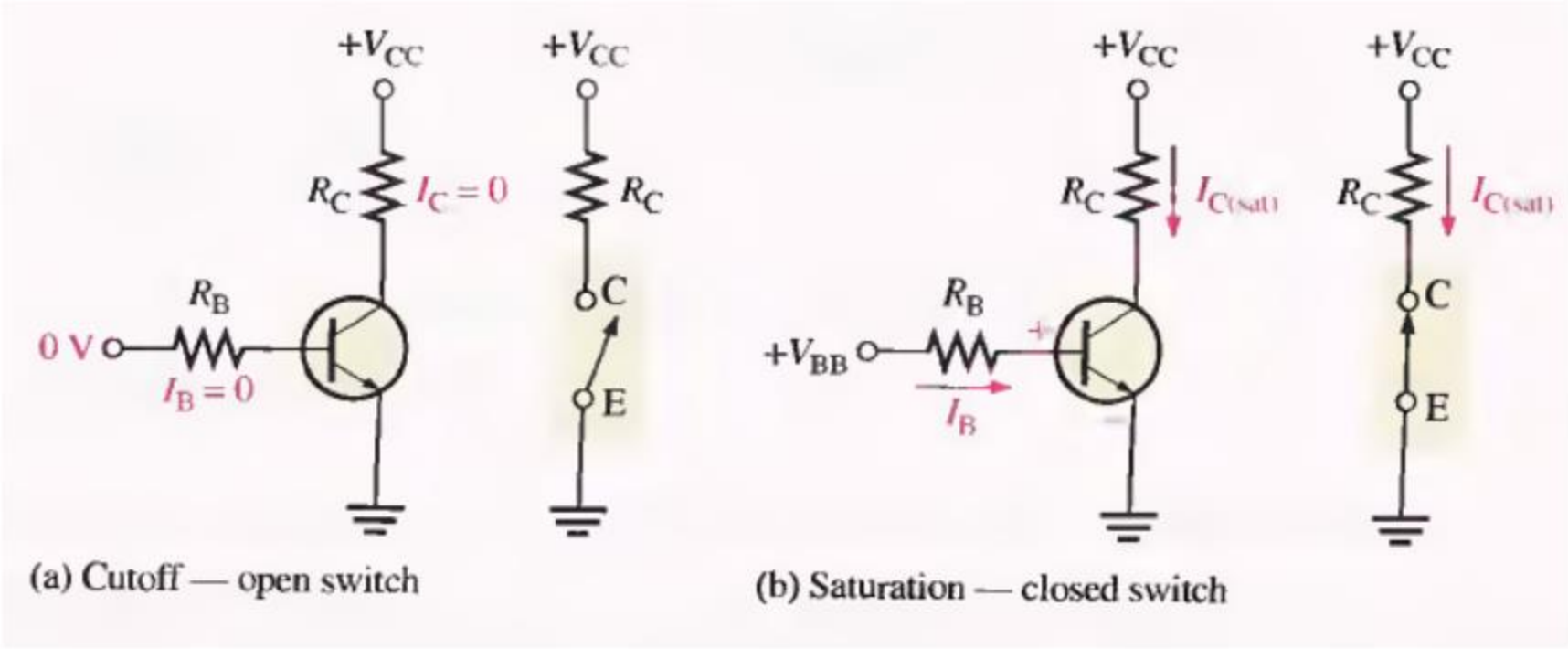


Fig. Ideal switching action of a transistor.

- **In part (a). the transistor is in the cutoff region because the base-emitter junction is not forward- biased.**
- **In this condition, the switch is ideally open, between collector and emitter, as indicated by the switch equivalent.**
- **In part (b), the transistor is in the saturation region because the base-emitter junction and the base-collector junction are forward-biased and the base current is made large enough to cause the collector current to reach its saturation value.**
- **In this condition, there is, ideally, a short circuit between collector and emitter, as indicated by the switch equivalent.**

A Simple Application of a Transistor Switch

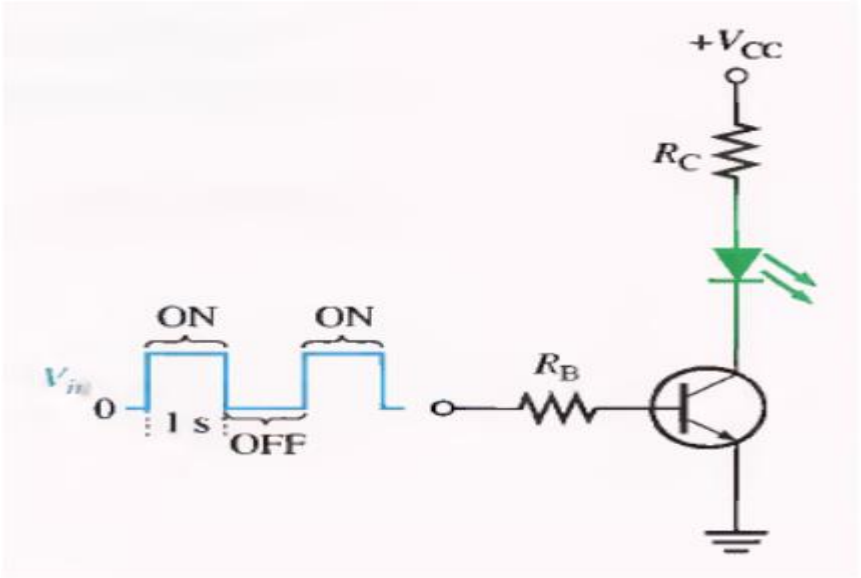
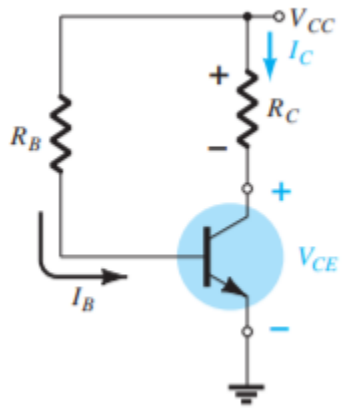


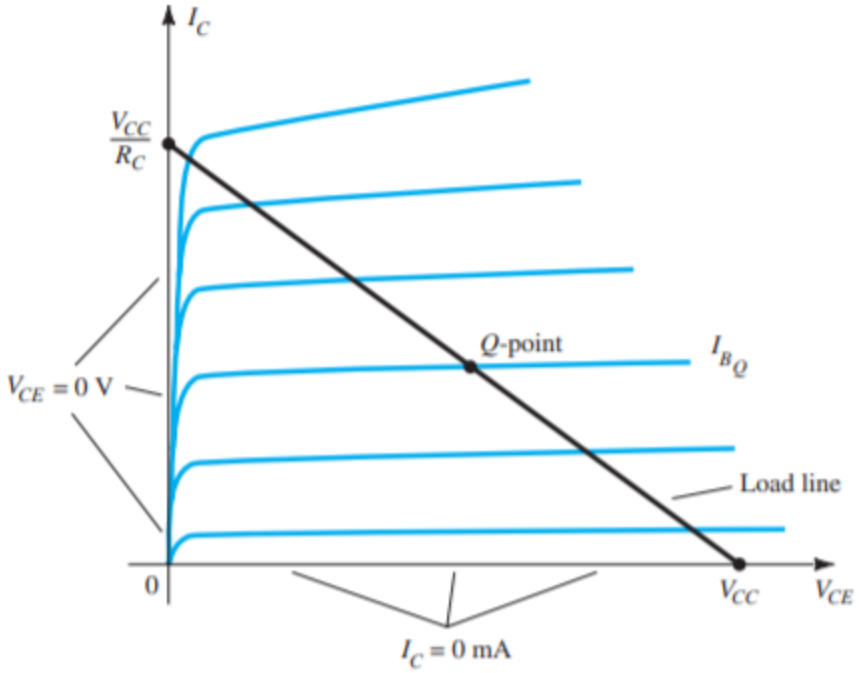
Fig. A transistor used to switch an LED on and off.

- When the square wave is at 0V, the transistor is in cutoff; and since there is no collector current, the LED does not emit light.
- When the square wave goes to its high level, the transistor saturates.
- This forward-biases the LED, and the resulting collector current through the LED causes it to emit light. Thus, the LED is on for 1s and off for 1s.

Biasing techniques

The DC- operating point





$$V_{CE} = V_{CC} - V_{RC}$$

$$V_{RC} = V_{CC} - V_{CE}$$

$$I_C R_C = V_{CC} - V_{CE}$$

$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$

When $V_{CE} = 0V$

$$I_{C(sat)} = \frac{V_{CC}}{R_C} \mid V_{CE} = 0$$ saturation

When $I_C = 0mA$

$V_{CE} = V_{CC} \mid I_C = 0mA$ cut-off

EX-

Determine the Q-point for the circuit in Figure below. Find the maximum peak value of base current for linear operation. Assume $\beta_{DC} = 200$.

